COMPSCI5079(M)
Cryptography & Secure Development

Dr. Nguyen Truong

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Who I am

Hanoi University of Science and Technology, Vietnam (BSc)

Pohang University of Science and Technology, South Korea (MSc)

DASAN Networks Corporation, South Korea (Software Engineer)

Liverpool John Moores University, Ph.D (2015-2018)

Imperial College London, Research Associate (2018-2022)

University of Glasgow, Lecturer (2022-onward)

Who I am

My research interest is Cyber Security, particularly *Data Privacy, Federated Learning, Blockchains and Decentralised Systems*.

- **Data Privacy**: *"GDPR-compliant personal data management: A blockchain-based solution"*, **NB Truong**, K Sun, GM Lee, Y Guo, *IEEE Transactions on Information Forensics and Security, 1746-1761, 2020*
- **Federated Learning**: *"Privacy Preservation in Federated Learning: An insightful survey from the GDPR Perspective"*, **NB Truong**, K Sun, Siyao Wang, Florian Guitton, Y Guo, *Elsevier Computers & Security (COSE) 110 (2021)*,
- **Blockchain and Bitcoin**: *"Strengthening the Blockchain-based Internet of Value with Trust", NB Truong, TW Um, B Zhou, GM Lee, IEEE International Conference on Communication (ICC) 2018*
- **Decentralised Systems**: *"A blockchain-based trust system for decentralised applications: When trustless needs trust"*, **N Truong**, GM Lee, K Sun, F Guitton, YK Guo, *Future Generation Computer Systems 124, 68-79, 2021*

Course Overview

This M.Sc. course covers two main goals:

(1) Encryption and Decryption algorithms, and
(2) How to utilise the algorithms in developing secure applications.

- The first part of this course focuses on understanding a variety of encryption and other cryptographic algorithms & schemes (2/3).

- The second part of this course focuses on developing secure applications (1/3). Demonstration applications are written in JAVA.

Course Overview

Coverage on how those Encryption and other cryptographic schemes are utilised is provided in:

- Cyber Security Fundamentals.

More specialised cyber security courses are:

- Enterprise Cyber Security
- Cyber Security Forensics
- Human Centred Security
- Safety Critical Systems

Course Structure

The course structure consists of 5 main sessions:

1. Fundamental Foundation for Cryptography
2. Symmetric Encryption schemes
   - Traditional Encryption
   - Modern Encryption schemes
   - Message digests, random numbers, multiple-key encryption
3. Public-key Encryption/Asymmetric key encryption
   - Diffie-Hellman, RSA
   - Elliptic Curve Cryptography
   - Crypto-currencies (e.g., Bitcoin, Ethereum) and Blockchain
4. Coding and Security
5. Secure Development

Textbooks

Some books are very useful for the course

1. *Applied Cryptography: Bruce Schneier.*

   Covers the cryptography part of the course.
2. *Bitcoin and Cryptocurrency Technologies: Narayanan, Bonneau, Felten, Miller, Goldfelder.*

   Bitcoin and Blockchain.
3. *Security Engineering: Ross Anderson.*

   The secure development part of the course.

Course Aims and Intended Learning Outcomes (ILOs)

Aims

1. To develop student's knowledge of cryptographic algorithms, how they can be attacked and how to evaluate how secure they are.
2. To develop student's practical skills in developing secure systems.

ILOs

1. Explain basic cryptographic algorithms, how they can be attacked and evaluate how secure they are.
2. Demonstrate an advanced understanding of a range of specialist algorithms, explaining when they are useful.
3. Produce a program that uses a standard cryptographic library to solve a security problem.
4. Critically compare and contrast many ways of developing secure systems.

Course Assessment

Assessment of the course:

- **80% Exam**
- **20% Coursework**
  - You will be given an individual set of encrypted messages and will have to find the plaintext using several different techniques.

Some bonus points for students who are doing well and showing their understanding and contribution during the lectures.

How to do well

Attend all the lectures and do the quizzes

- **Learn as you go**
  - It is too difficult to try and learn the course just before the exam.
  - Lots of examples and instructions on the Internet. Try to learn proactively.
- **Do the coursework**
  - Most marks are in the exam; however, doing the coursework helps you understand more about the lectures.